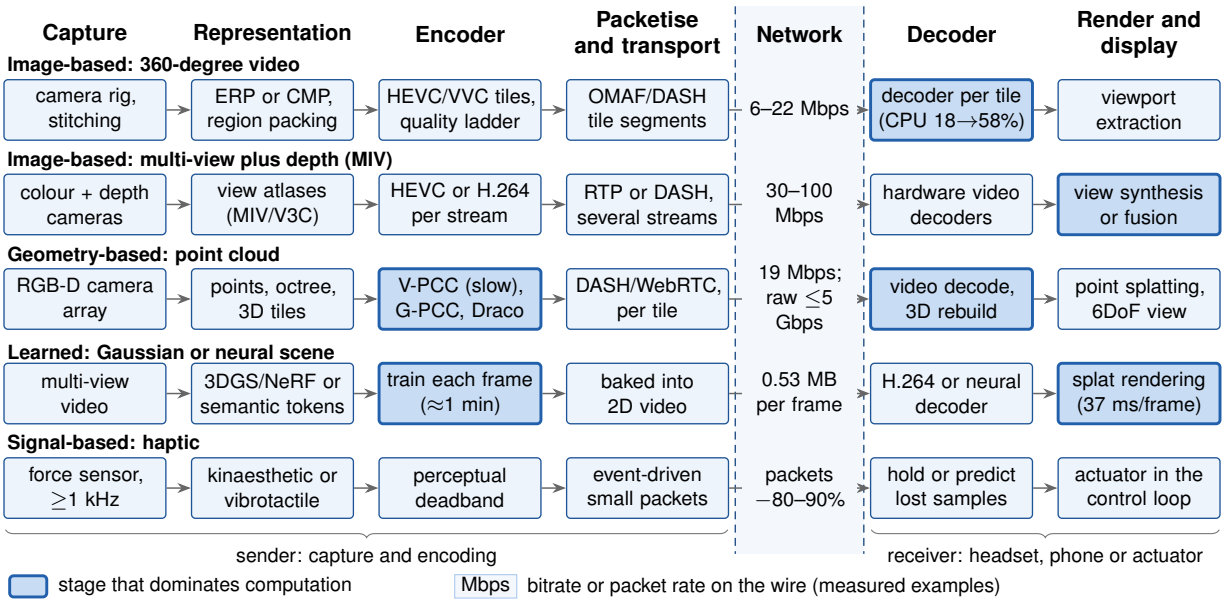


Figure 1: Transmission chains from capture to display for one or two representations of each family. No chain is cheap at both ends: a low rate on the wire is bought with encoder delay (V-PCC, per-frame training) or with computation on the receiver (tile decoding, view synthesis, splat rendering). Figures: 360-degree video [1, 2], multi-view streams [3], point clouds [4, 5], Gaussian video [6], deadband coding [7].

1 What Is Sent: Representing and Encoding Immersive Content

Every networked XR system must first decide what its content is: a sphere of pixels, a set of camera views, a cloud of points, the parameters of a learned scene model or a stream of force samples. The design question is which representation travels and how it is compressed. The choice is made before any network algorithm acts, and it fixes four quantities that every later function inherits. The *bitrate floor* is the rate below which quality collapses, and it sets what the network must supply (**??**). The *encoder delay* is spent before the first bit leaves the sender and counts against the interaction budgets of **????**. The *decoder cost* lands on a battery-powered headset or phone. *Viewport dependence* decides whether the stream can be cut down to what the user will see, and therefore whether prediction and prefetching (**??**) have anything to act on.

We group representations into four families by what they send (Table 1). *Image-based* representations send what cameras see and reuse the 2D video chain. *Geometry-based* representations send the shape of the scene as points or meshes. *Learned* representations send a model of the scene, or a description of its meaning, and let the receiver regenerate the content. *Signal-based* representations send a physical signal, force or light, sampled by the rules of the sense that receives it. Figure 1 follows a chain from each family from capture to display. Adaptation and caching are treated in **??**, haptic transport and holoportation in **??**.

1.1 Image-based: send what a camera sees

Image-based representations keep the 2D video chain intact and inherit its hardware encoders and decoders. Their questions sit at the two ends of that chain: how the scene is mapped onto rectangles, and how many decoders the client can afford.

Projected 360-degree video assumes a viewer who turns the head but stays in place (3DoF), and wins when that holds. The established result is that concentrating bits on the viewport saves bitrate, but less than it saves pixels: Oculus's offset cube used fewer than half the pixels of an equirectangular frame for similar view quality, yet saved only 5.6–16.4% of bitrate [1]. The delivery unit is contested. Viewport-specific versions need no reassembly but multiply storage. Tiles share one encode but cost 7.3–10.4% in coding efficiency [2], up to 37.3% for small tiles at low resolution [8], and the client pays for them: Flare raised phone CPU load from 18% to 58% [2], and streaming took 28.5% of headset power [9], both on 2017–2018 hardware. Gaze is the stronger lever: gaze-based quantisation saved 40–47% in cloud VR gaming in a subjective test [10], and gaze-contingent compression was five times more effective when motion-to-photon (MTP) latency stayed below 15 ms [11]. Foveation is thus a

Table 1: Representation families and where their cost lands. Figures are examples measured on the cited systems under different conditions; they locate each cost but are not matched operating points.

Family	Idea	Cost on wire	Cost at sender	Cost at receiver	Maturity
Image	360-degree video: sphere projected to rectangles, bits on the viewport	6–22 Mbps	low; tiles lose 7–10% efficiency	a decoder per tile, CPU 18→58%	standard (OMAF), deployed
Image	Multi-view plus depth (MIV): camera views, new views synthesised	30–100 Mbps	low; hardware video encoders	several decoders and view synthesis	standard; rarely tested on networks
Geometry	Point clouds (V-PCC, G-PCC, Draco)	19 Mbps live; raw up to 5 Gbps	V-PCC hours per second of content; G-PCC, Draco fast	3D rebuild; 30 fps on phones needs inter-frame coding	standard; on-demand prototypes
Geometry	Meshes: surfaces from fused depth	14 Mbps mean (reconstructed scene)	reconstruction	rasterisation	static codecs benchmarked
Learned	Neural scene (NeRF, 3DGS): model parameters per frame	0.53 MB per frame	training ≈1 min per frame	splat rendering, 37 ms per frame	research prototypes
Learned	Semantic and generative: meaning, not pixels	93–99.9% less	inference, shared knowledge base	generative inference; 13.7 s end to end	simulation, one 5G prototype
Signal	Haptic: samples sent only above a perceptual threshold	80–90% fewer packets	negligible, delay-free	hold or predict missing samples	standard (IEEE 1918.1.1)
Signal	Holographic: light field or phase pattern	6.63 TBps by arithmetic; 173 Mbps measured	V-PCC 11–42 min per second	display not characterised	requirement tables

coding gain that **??** pays for in latency.

Multi-view plus depth sends several colour and depth streams and lets the receiver synthesise the view, which gives limited six degrees of freedom (6DoF) with ordinary video codecs. It assumes the receiver can afford view synthesis and the viewer stays within the span the cameras covered. It wins in live, two-way systems because it needs no slow 3D encoder. Project Starline sends four colour and three depth H.265 streams at 30–100 Mbit/s and fuses them at the receiver, because real-time mesh or point-cloud compression was not possible at its resolution [3]; MetaStream likewise codes RGB-D as 2D video [12]. MPEG Immersive Video (MIV) standardises this path, with atlas packing [13] and view synthesis at the decoder [14]. Practice and evidence diverge: deployed telepresence uses this family, yet no study has streamed MIV to users over a real network, and light-field delivery has been tested only offline [15].

1.2 Geometry-based: send the shape

Geometry-based representations send the scene itself, so the receiver can render any viewpoint. The price is weight. Raw point clouds need 720 Mbit/s at 200K points and 30 fps [5], and about 5 Gbit/s at one million points [4], one to three orders of magnitude more than 2D video.

Point-cloud codecs trade compression ratio against speed, and the trade is established. V-PCC projects patches onto planes and reuses video codecs; it reaches lossy ratios of 100–500 but took about 4 hours to encode 30 frames of one object [16]. G-PCC and Draco code geometry in 3D, run fast and compress less, so live systems avoid V-PCC. In VR2Gather, the MPEG Anchor codec cut 281.8 Mbit/s to 19.0 Mbit/s but raised latency from 40.6 to 278.7 ms, and parallel tiles brought it back to 90.6 ms [4]: coding, not transit, set the delay. Tuning standard codecs yields single-digit gains (2–6% BD-rate from V-PCC rate control [17]); larger gains come from changing the representation, as when 3D sub-sampling cut bitrate by up to 48.5% at equal quality [18] and 3D patch coding matched V-PCC's ratio with real-time decoding [19]. The receiver is the second constraint: Hermes reached 30 fps on phones only by exploiting inter-frame correlation [20]. No codec wins everywhere; a measurement

of Draco, G-PCC, V-PCC and two neural codecs found that the best choice depends on content and viewing behaviour [21].

Geometry's advantage is a richer viewport dependence: points can be culled by visibility, occlusion and distance, not only by direction. ViVo saved 40% of bandwidth on average with SSIM above 0.99, but only 10% for phone users on the same content [5], and gaze-driven delivery on a HoloLens 2 saved up to 67% [22]. Meshes are the less explored member: static mesh codecs have been benchmarked for rate and runtime [23], and a reconstructed mesh for robot teleoperation needed 14 Mbit/s on average [24].

1.3 Learned: send a model or a meaning

Learned representations replace explicit content with something the receiver must regenerate. They promise the largest savings and move the most work to the two ends.

Neural scene representations (NeRF, 3D Gaussian splatting) send model parameters. The most deployable design reuses existing hardware: V^3 bakes dynamic Gaussians into 2D videos for H.264 decoders and reaches 32.97 dB PSNR at 0.532 MB per frame and 27 fps on a phone [6]. That rate is not small: sustained at 30 fps it is about 128 Mbit/s, the range of multi-view video, so the gain is rendering freedom rather than bitrate. Training takes about a minute per frame on a desktop GPU, which rules out live content, and download was measured only on an internal network. Layered Gaussians [25], variable-bitrate NeRF video [26] and splat streaming at 30 fps on phones [27] point towards adaptation; error control and quality assessment remain open [28].

Semantic and generative codecs send meaning and assume that sender and receiver share a knowledge base or a generative model. They report the largest savings and the weakest comparisons. WiserVR sent 1.81 Mbit instead of 28.3 Mbit for 20 frames, in an AWGN simulation that excluded the delay of sharing the knowledge base [29]; MagicStream reports up to 1195$\times$ less bandwidth [30]. The one generative prototype timed end to end cut bandwidth by 99.93% but raised latency from 980 ms to 13.66 s [31]. Each paper defines fidelity differently, task metrics can disagree with perceived realism [32], and the baseline is usually BPG with LDPC rather than HEVC or VVC. An order-of-magnitude saving is plausible; its price in inference delay and device energy is assumed rather than measured.

1.4 Signal-based: send the physical signal

Signal-based representations code a physical quantity by the limits of the sense that receives it. Haptic and holographic content sit at opposite ends of the rate scale.

Haptic signals need little rate and strict timing. Kinaesthetic samples travel at 1 kHz or more to keep the control loop stable, and perceptual deadband coding sends a sample only when its change exceeds a Weber fraction (7–15% for force), removing 80–90% of packets without perceptible loss. Two independent implementations of the IEEE 1918.1.1 kinaesthetic codec reproduced the same rates and errors [7], an established result, though for one contact point. Vibrotactile coding is open-loop and may use block transforms with tens of milliseconds of delay [33]. Learned multichannel codecs report cuts of 80–92%, but against each other rather than a standard [34, 35]. Deadband coding also makes traffic bursty, unlike the periodic 1 ms flow that transport studies assume.

Holographic content has no agreed representation, and its requirement figures depend on an unstated one. Akyildiz and Guo define a hologram as a recorded light field, then adopt point clouds as the working form; their light-field display example needs 221 GB per frame, or 6.63 TBps at 30 fps [36]. Measured systems used 5–100 Mbps for compressed point clouds over live 5G [37] and 173 Mbps for full-duplex light-field telepresence [38]. The figures do not estimate one quantity: one is uncompressed display arithmetic, the other compressed capture. For live use, codec delay binds first: V-PCC needs 11–42 minutes to encode one second [39]. Optical holograms are only now becoming coded media [40].

The choice moves cost; it does not remove it. No representation is cheap at both ends. Image-based content is light on the sender and heavy on the wire and the decoders. Geometry cuts the wire at the price of encoder delay. Learned representations cut it further and move the work into training and inference. Choosing a representation is therefore choosing where to spend: on the network, the sender's encoder or the receiver's processor, and the choice cannot be made alone. Remote rendering (**??**) collapses any representation back into 2D video at the edge, trading decoder cost for

MTP latency. Viewport dependence bounds what prediction can save (**??**), and encoder delay decides which representations two-way telepresence can use (**??**). The ratio–speed trade within each family is established; that new representations win across families is assumed, since none has been compared with the others at matched quality.

Open problems

P1. Which representation minimises end-to-end cost for a given scene? Point clouds, meshes, multi-view video and Gaussian splats have never been compared on the same content at matched perceived quality. Encoding shared scenes in all four and streaming them over replayed 5G and Wi-Fi traces would measure bitrate, encoder delay and decoder load together.

P2. Do semantic and generative savings survive latency? Reported savings come from AWGN simulations with weak baselines. Comparing semantic codecs with HEVC, VVC and V-PCC at matched rated quality on a real 5G link, timing encoder and decoder inference, would show whether they suit interactive use.

P3. How should new representations be judged? No rating data exist for compressed Gaussian, neural or generative reconstructions viewed in a headset, so PSNR and LPIPS are unvalidated for their artefacts. A 6DoF rating study across codecs and rates would provide ground truth.

P4. Can codecs track a mobile link? Rate control is tuned at fixed rates, and neural codecs lack bitrate ladders and loss handling. Rate control driven by measured capacity, judged by frame-deadline misses, would test whether codec gains hold on cellular links.

P5. What does a hologram actually cost to send? Requirement tables rest on display arithmetic, not on captured content. Coding one scene as point clouds, splats, light fields and phase holograms and recording the traffic would replace assumed figures with measured ones.

P6. How do coded haptic streams behave under real loss? Haptic codecs are evaluated on one-off recordings with one contact point. Shared multi-point recordings and a user study of deadband and vibrotactile codecs over replayed network impairments would test perceptual transparency where it matters.

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